UNIT

V

ENGINEERING PHYSICS • DIGITAL NOTES

Semiconductors

Energy bands • Classification of solids • Intrinsic semiconductors • Extrinsic (n and p type) • Fermi level • Drift and diffusion • Einstein relation • Hall effect and applications

QUICK SUMMARY • What you will learn

- How energy bands form when atoms come together and why that sorts solids into conductors, semiconductors and insulators
- How to find carrier density, Fermi level and conductivity in intrinsic and doped semiconductors
- How currents flow (drift, diffusion) and how the Hall effect tells us the type and density of carriers

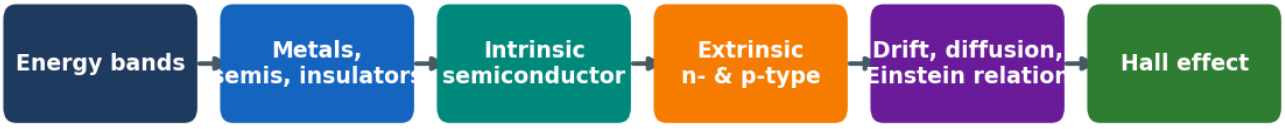


Fig. 1: Roadmap of Unit V

1 Formation of Energy Bands

An isolated atom has sharp, discrete energy levels. When N identical atoms are brought together, the electron clouds overlap, and the **Pauli exclusion principle** forbids two electrons from having the same state. Each atomic level therefore **splits into N closely spaced levels**, which merge into a **continuous band** ($N \approx 10^{23}$).

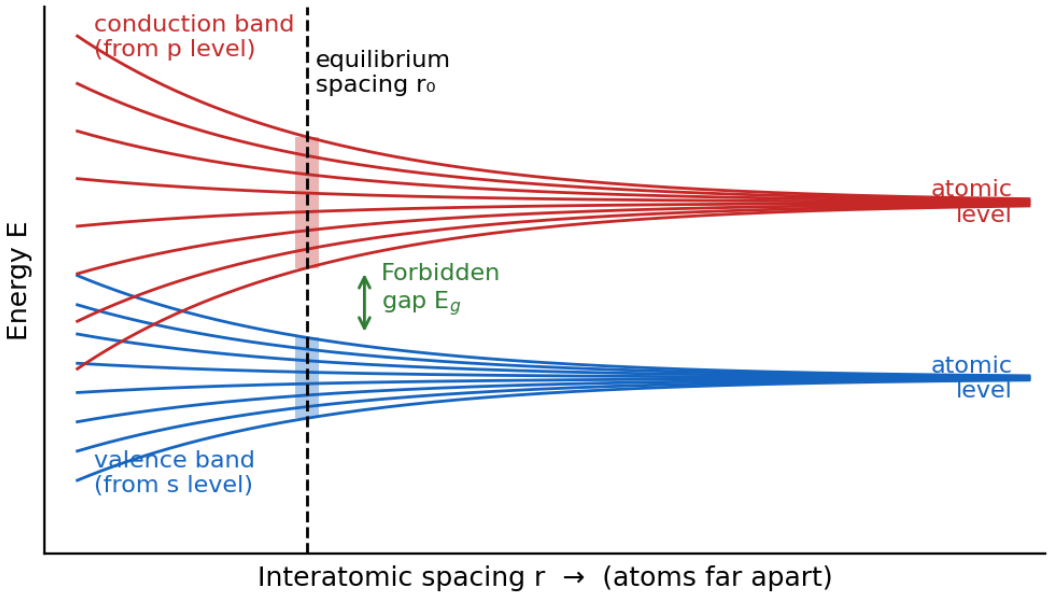


Fig. 2: Splitting of atomic levels into bands as the interatomic spacing decreases

- Outer (valence) electron levels spread most (strong overlap); inner levels stay narrow.
- **Valence band (VB):** highest band filled with valence electrons. **Conduction band (CB):** next higher band, empty or partly filled.
- **Forbidden gap (E_g):** energy range between the top of VB and bottom of CB with no allowed levels.
- The band picture is the key that sorts all solids by how easily electrons can reach empty states.

DEFINITION • Hole and effective mass

A **hole** is the vacancy left in the almost full valence band. It behaves as a particle of charge +e. The **effective mass m*** packages the effect of the crystal's periodic potential on a carrier: $m^* = \hbar^2/(d^2E/dk^2)$. Near band edges the carrier then behaves like a free particle of mass m*.

REMEMBER • Kronig–Penney idea (qualitative)
An electron in the periodic potential of a lattice has allowed energy bands separated by forbidden gaps because waves with k = nπ/a are Bragg-reflected and form standing waves with two different energies, leaving a gap between them.

2 Classification of Crystalline Solids

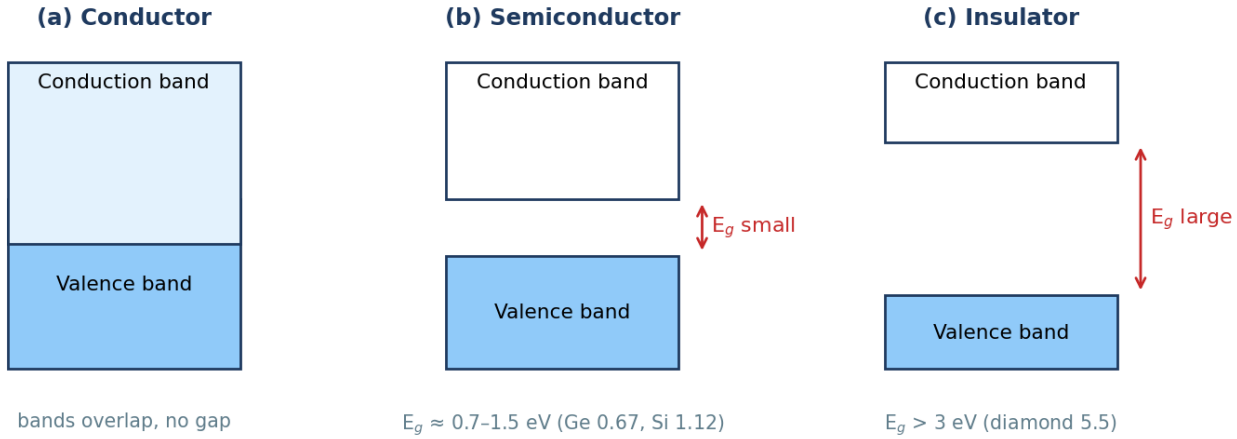


Fig. 3: Band diagrams of a conductor, a semiconductor and an insulator

Property	Conductor (metal)	Semiconductor	Insulator
Band gap E_g	None (bands overlap or CB part filled)	≈ 0.1–3 eV	> 3 eV
Resistivity (Ω·m)	10^{-8}–10^{-6}	10^{-4}–10^{6}	10^{8}–10^{16}
Free carriers at 300 K	Very many (~10^{28} m^{-3})	Few (~10^{16}–10^{19} m^{-3})	Almost none
Effect of rising T	ρ increases (more scattering)	ρ decreases (more carriers)	ρ decreases, but stays high
Examples	Cu, Al, Ag	Si (1.12 eV), Ge (0.67 eV), GaAs (1.43 eV)	Diamond (5.5 eV), glass, mica

KEY IDEA • Temperature coefficient
At 0 K a pure semiconductor is an insulator: VB full, CB empty. Thermal energy kT (0.026 eV at 300 K) lifts a few electrons across E_g leaving **holes** behind, which behave as positive carriers.

3 Intrinsic Semiconductors

DEFINITION • Intrinsic semiconductor
A pure semiconductor (no impurities) in which electrons in the CB and holes in the VB are produced only by thermal generation of electron\u2013hole pairs, so n = p = n_i.

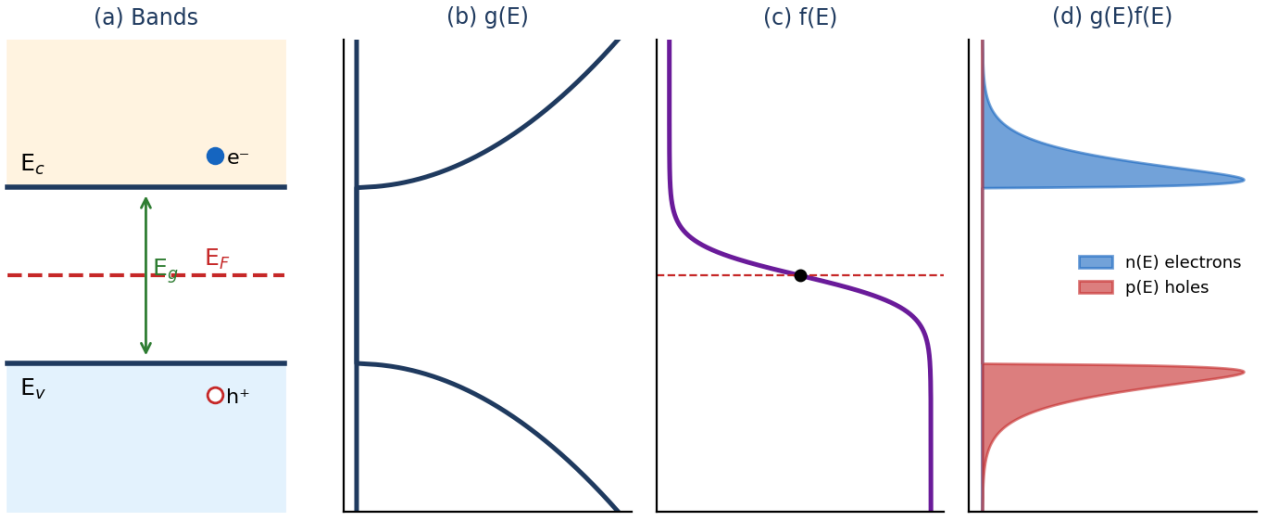


Fig. 4: Bands, density of states g(E), Fermi function f(E) and the resulting carrier distributions

Density of charge carriers

Electron density in the CB: $n = \int g_c(E)\, f(E)\, dE$ from E_c upward, with $g_c(E) = (4\pi/h^3)(2m_e^*)^{3/2}(E - E_c)^{1/2}$. Since $E - E_F \gg kT$ (non-degenerate), $f(E) \approx e^{-(E-E_F)/kT}$. Integrating:

Electrons in CB	$n = N_c\, e^{-(E_c - E_F)/kT}, \qquad N_c = 2\left(\frac{2\pi m_e^* kT}{h^2}\right)^{3/2}$
Holes in VB (probability $1 - f$)	$p = N_v\, e^{-(E_F - E_v)/kT}, \qquad N_v = 2\left(\frac{2\pi m_h^* kT}{h^2}\right)^{3/2}$
Intrinsic: $n = p = n_i$	$n_i = 2\left(\frac{2\pi kT}{h^2}\right)^{3/2}(m_e^* m_h^*)^{3/4}\, e^{-E_g/2kT}$
Law of mass action	$n\,p = n_i^2 = N_c N_v\, e^{-E_g/kT}$

Fermi level

KEY IDEA • Derivation in three lines

- $n = p \rightarrow N_c\, e^{-(E_c - E_F)/kT} = N_v\, e^{-(E_F - E_v)/kT}$.
- Take logarithms: $2E_F = E_c + E_v + kT \ln(N_v/N_c)$.
- $N_v/N_c = (m_h^*/m_e^*)^{3/2} \rightarrow E_F = (E_c + E_v)/2 + (3kT/4)\ln(m_h^*/m_e^*)$.

Setting $n = p$ and solving for E_F:

Intrinsic Fermi level	$E_F = \frac{E_c + E_v}{2} + \frac{3kT}{4}\ln\frac{m_h^*}{m_e^*}$

If $m_h^* \approx m_e^*$ (or at T = 0 K), E_F lies in the **middle of the gap**, at $E_g/2$ above the VB. Because m_h^* is a bit larger than m_e^*, E_F drifts slightly upward with T.

Electrical conductivity

Conductivity	$\sigma = ne\mu_e + pe\mu_h = n_i e\,(\mu_e + \mu_h) \;\propto\; T^{3/2} e^{-E_g/2kT}$
Straight-line test	$\ln\sigma = \text{const} - \frac{E_g}{2k}\cdot\frac{1}{T} \;\Rightarrow\; \text{slope} = -\frac{E_g}{2k}$

EXAMPLE • Numerical

Germanium at 300 K: $n_i = 2.4 \times 10^{19}\ m^{-3}$, $\mu_e = 0.39$ and $\mu_h = 0.19\ m^2/Vs$. $\sigma = n_i e(\mu_e + \mu_h) = 2.4\times10^{19} \times 1.6\times10^{-19} \times 0.58 =$ **2.2 (Ωm)$^{-1}$**.

APPLICATIONS • Why semiconductors matter

- Diodes, transistors and ICs: controlled conductivity by doping.
- Solar cells and photodetectors: photons of energy ≥ E_g create electron–hole pairs.
- LEDs and laser diodes: direct-gap materials (GaAs, GaN) emit light on recombination; Si is indirect and a poor emitter.
- Thermistors, Hall sensors, gas sensors.

4 Extrinsic Semiconductors

Doping adds a controlled trace (1 part in 10^6–10^9) of impurity atoms. Result: an **extrinsic semiconductor** whose conductivity is set mostly by the dopant.

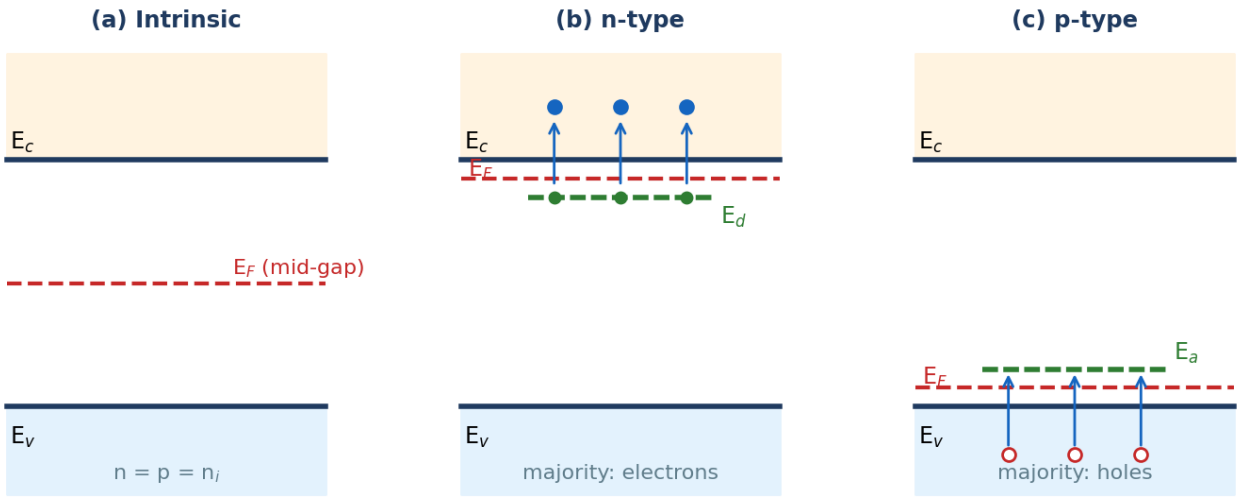


Fig. 5: Energy bands of intrinsic, n-type and p-type semiconductors

	n-type	p-type
Dopant	Pentavalent donor (P, As, Sb)	Trivalent acceptor (B, Al, Ga, In)
Extra level	Donor level E_d just **below** E_c (~0.01–0.05 eV)	Acceptor level E_a just **above** E_v
What happens	Fifth electron is easily freed to the CB; donor becomes a fixed positive ion	Accepts an electron from VB; leaves a mobile hole and a fixed negative ion
Majority / minority	Electrons / holes	Holes / electrons
At 0 K, E_F	Midway between E_d and E_c	Midway between E_v and E_a

Feature	Intrinsic	Extrinsic
Purity	Pure crystal	Doped (10^{-6}–10^{-9})
Carriers	$n = p = n_i$	$n \neq p$; majority + minority
Conductivity	Low, strongly T-dependent	Higher and nearly T-independent in the working range
Fermi level	Mid-gap	Near E_c (n) or E_v (p)
Use	Reference and theory	All devices

Carrier density in extrinsic semiconductors

n-type (low T, $\Delta E = E^c - E^d$)

$$n = (2N_d)^{1/2}\left(\frac{2\pi m_e^* kT}{h^2}\right)^{3/4} e^{-\Delta E/2kT}$$

p-type (low T, $\Delta E = E_a - E_v$)

$$p = (2N_a)^{1/2}\left(\frac{2\pi m_h^* kT}{h^2}\right)^{3/4} e^{-\Delta E/2kT}$$

Mass action (any T)

$$np = n_i^2 \Rightarrow p_n = \frac{n_i^2}{N_d}, \quad n_p = \frac{n_i^2}{N_a}$$

Conductivity

$$\sigma_n \approx N_d e\mu_e, \qquad \sigma_p \approx N_a e\mu_h$$

Dependence of Fermi energy on concentration and temperature

n-type, near 0 K

$$E_F = \frac{E_c + E_d}{2} + \frac{kT}{2}\ln\frac{N_d}{2(2\pi m_e^* kT/h^2)^{3/2}}$$

p-type, near 0 K

$$E_F = \frac{E_v + E_a}{2} - \frac{kT}{2}\ln\frac{N_a}{2(2\pi m_h^* kT/h^2)^{3/2}}$$

After full ionisation

$$E_F - E_i = kT\ln\frac{N_d}{n_i}\ \text{(n-type)}, \quad E_i - E_F = kT\ln\frac{N_a}{n_i}\ \text{(p-type)}$$

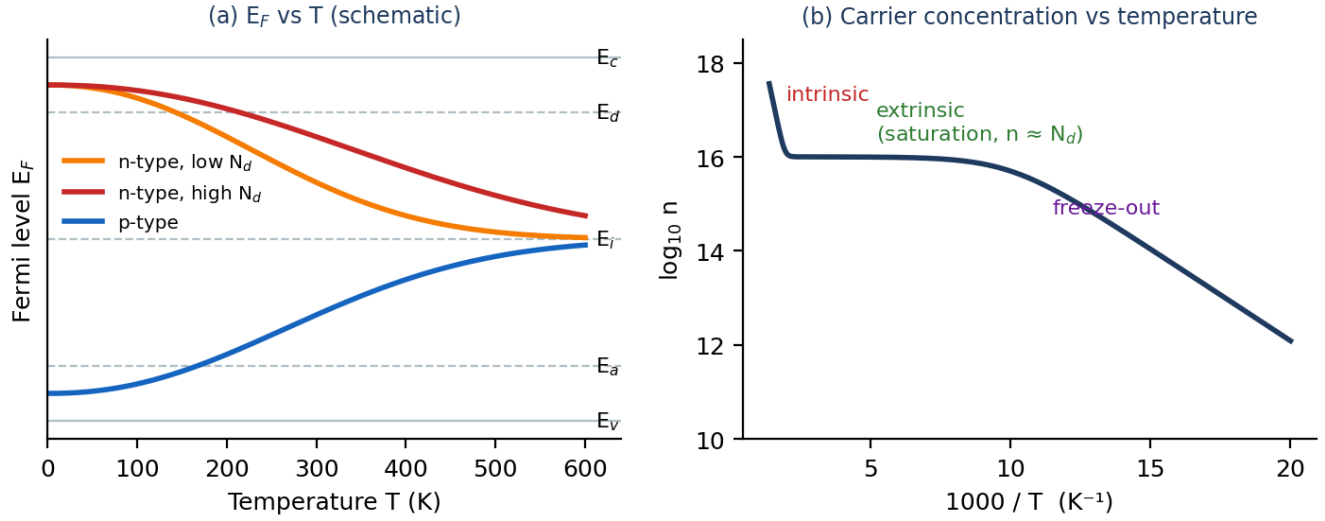


Fig. 6: Fermi level versus temperature and carrier concentration versus 1/T (schematic)

- **Concentration:** more N_d pushes E_F closer to E_c (n-type); more N_a pushes it closer to E_v (p-type).
- **Temperature:** as T rises, E_F moves from near the band edge toward the **intrinsic level E_i** (mid-gap). Heavier doping needs a higher temperature before the semiconductor turns intrinsic.
- Three regions in n(T): **freeze-out** (low T, donors not all ionised) → **saturation** ($n \approx N_d$, device range) → **intrinsic** (high T, $n \approx n_i$).

EXAMPLE • Numerical
Silicon doped with $N_d = 10^{22}$ m^{-3} at 300 K ($n_i = 1.5 \times 10^{16}$ m^{-3}): $n \approx 10^{22}$ m^{-3}; $p = n_i^2/N_d$ = **2.25×10^{10} m^{-3}**; $E_F - E_i = kT \ln(N_d/n_i) = 0.0259 \times 13.4$ = **0.347 eV** above mid-gap.

5 Drift and Diffusion Currents; Einstein's Relation

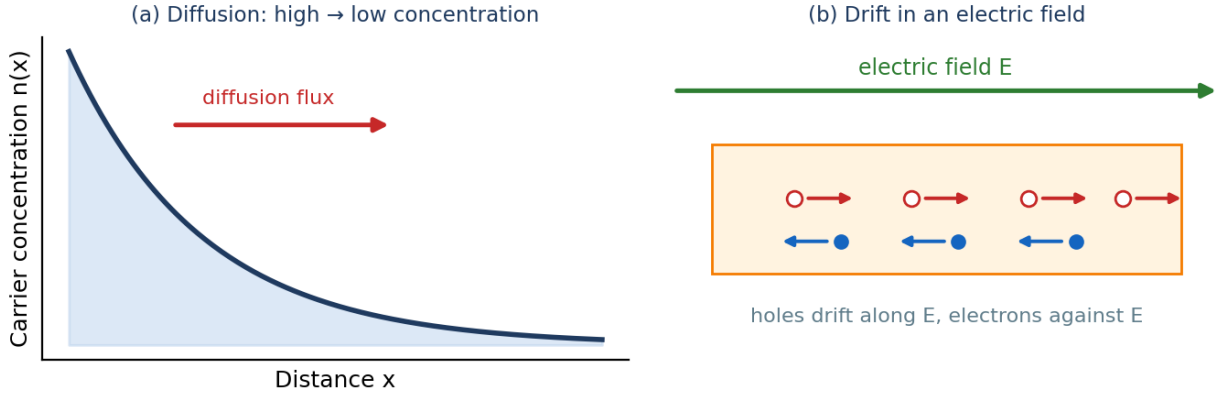


Fig. 7: Diffusion (concentration gradient) and drift (electric field)

	Drift current	Diffusion current
Cause	Electric field E	Concentration gradient dn/dx or dp/dx
Mobility / diffusion	Mobility $\mu = v_d/E$ (m²/Vs)	Diffusion coefficient D (m²/s)
Electrons	$J_n = n\,e\,\mu_n\,E$	$J_n = e\,D_n\,(dn/dx)$
Holes	$J_p = p\,e\,\mu_p\,E$	$J_p = -e\,D_p\,(dp/dx)$

Total current density $$J = J_n + J_p = e(n\mu_n + p\mu_p)E + e\left(D_n\frac{dn}{dx} - D_p\frac{dp}{dx}\right)$$

Einstein relation $$\frac{D_n}{\mu_n} = \frac{D_p}{\mu_p} = \frac{kT}{e}$$

Diffusion length $$L = \sqrt{D\tau}$$

KEY IDEA • Proof of the Einstein relation (idea)

- In equilibrium the net current is zero: $n\mu E + D\, dn/dx = 0$.
- Boltzmann distribution in potential V(x): $n = n_0\, e^{eV/kT} \rightarrow dn/dx = -(e/kT)\, nE$.
- Substitute: $n\mu E - D(e/kT)nE = 0 \rightarrow$ **$D/\mu = kT/e$** (= 0.0259 V at 300 K).

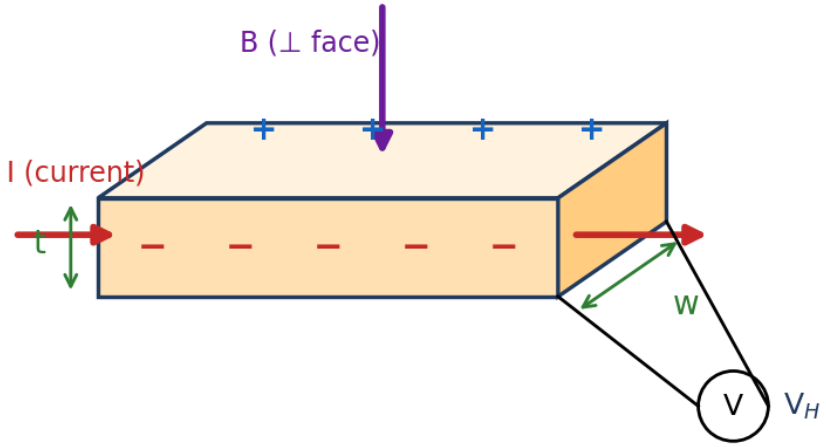
EXAMPLE • Numerical

In Si, $\mu_e = 0.135$ m²/Vs. $D_e = \mu kT/e = 0.135 \times 0.0259 =$ **3.5×10^{-3} m²/s**.

6 Hall Effect and its Applications

DEFINITION • Hall effect

When a current-carrying conductor or semiconductor is placed in a magnetic field perpendicular to the current, a transverse voltage (Hall voltage V_H) appears across the sample, perpendicular to both I and B.



n-type: electrons (drift opposite to I) are pushed to the front face by $F = -e(v \times B)$

Front face turns negative, back face positive → Hall voltage V_H across width w

Fig. 8: Hall effect in an n-type slab: current I along x, field B along z, Hall voltage along y

Carriers drifting with velocity v_d feel a sideways magnetic force Bev_d. They accumulate on one face until the electric Hall field E_H balances it: $eE_H = Bev_d$.

Hall field	$E_H = Bv_d = \frac{BJ}{ne} \quad (J = nev_d)$
Hall voltage (w = width, t = thickness, I = J w t)	$V_H = E_H w = \frac{IB}{net}$
Hall coefficient	$R_H = \frac{E_H}{JB} = \frac{V_H t}{IB} = -\frac{1}{ne}$ (n-type), $+\frac{1}{pe}$ (p-type)
Hall mobility	$\mu_H = \lvert R_H \rvert \sigma$

Sample	Majority carrier	Sign of R_H	Polarity of the front face (Fig.)
n-type	Electrons	Negative	Negative (electrons pile up)
p-type	Holes	Positive	Positive (holes pile up)
Metal (e.g. Cu)	Electrons	Negative, tiny (n ~ 10^{28})	V_H very small → poor Hall sensor

APPLICATIONS • Applications of the Hall effect

- **Type of semiconductor:** sign of V_H (or R_H) shows whether carriers are electrons (−) or holes (+).
- **Carrier concentration:** $n = 1/(e|R_H|)$.
- **Mobility:** $\mu = |R_H|\,\sigma$ using the measured conductivity.
- **Magnetic field measurement:** Hall probe (gaussmeter) since $V_H \propto B$.
- Hall sensors in position/speed sensing, current sensors, brushless motors, power measurement ($V_H \propto$ product of I and B).

EXAMPLE • Numerical

n-type sample, $n = 10^{22}$ m^{-3}, t = 0.5 mm, I = 10 mA, B = 0.5 T: $V_H = IB/(\text{net}) = (10^{-2} \times 0.5)/(10^{22} \times 1.6\times10^{-19} \times 5\times10^{-4}) =$ **6.25 mV**.
Given $R_H = 3.66\times10^{-4}$ m³/C and $\sigma = 112\ (\Omega m)^{-1}$: $n = 1/(eR_H) =$ **1.7×10^{22} m^{-3}**, $\mu = R_H\sigma =$ **0.041 m²/Vs**.

Mobility and temperature, direct and indirect gaps

Topic	Key point
Mobility vs T	Lattice scattering: $\mu \propto T^{-3/2}$ (dominant at high T). Impurity scattering: $\mu \propto T^{3/2}$ (dominant at low T). So μ peaks at some intermediate temperature.
Direct gap (GaAs, GaN)	CB minimum and VB maximum at the same k: electron–hole recombination gives a photon → LEDs and lasers.
Indirect gap (Si, Ge)	Minimum and maximum at different k: recombination needs a phonon too, so light emission is weak.
Compound semiconductors	III–V (GaAs, InP) and II–VI (ZnS, CdS) tune E_g and colour by composition.

EXAM CORNER • True or false?

- The Fermi level of an intrinsic semiconductor depends strongly on temperature. **False** (it stays near mid-gap).
- Doping an intrinsic semiconductor increases its conductivity. **True**.
- In an n-type semiconductor there are no holes. **False** ($p = n_i^2/N_d$, small but non-zero).
- Hall voltage is zero if B is parallel to I. **True** (no sideways force).

7 Worked Problems

EXAMPLE • 1. Intrinsic carriers

For Si (E_g = 1.12 eV) $n_i = 1.5\times10^{16}$ m^{-3} at 300 K. The ratio at 400 K is approximately $(400/300)^{3/2}\, e^{(E_g/2k)(1/300 - 1/400)} = 1.54 \times e^{6.5\times10^{3}\times8.33\times10^{-4}} = 1.54 \times e^{5.4} \approx$ **3.5×10^2**, so n_i(400 K) ≈ 5×10^{18} m^{-3}.

EXAMPLE • 2. Conductivity

n-type Si with $N_d = 10^{22}$ m^{-3}, μ_e = 0.135 m²/Vs: $\sigma \approx N_d e\mu_e = 10^{22} \times 1.6\times10^{-19} \times 0.135 =$ **216 $(\Omega m)^{-1}$**, about 5×10^5 times more than intrinsic Si (≈ 4.4×10^{-4}).

EXAMPLE • 3. Band gap from a photon

A GaAs LED emits λ = 870 nm: $E_g = 1240/870 =$ **1.43 eV**. Si (1.12 eV) absorbs light up to λ = 1240/1.12 = **1107 nm**.

EXAMPLE • 4. Fermi level of p-type

Si with $N_a = 10^{22}$ m^{-3} at 300 K: $E_i - E_F = kT \ln(N_a/n_i) =$ **0.347 eV**, so E_F is 0.347 eV below mid-gap (0.21 eV above E_v).

EXAMPLE • 5. Diffusion

Holes in Si with μ_h = 0.048 m²/Vs: $D_h = 0.048 \times 0.0259 =$ **1.24×10^{-3} m²/s**; with τ = 10 μs, $L_h = \sqrt{(D\tau)} =$ **1.1×10^{-4} m**.

Key terms at a glance

Term	Meaning
Generation / recombination	Creation / annihilation of an electron–hole pair
Majority / minority carriers	Carriers in excess / in deficit compared with the other type
Degenerate semiconductor	So heavily doped that E_F enters a band and it behaves like a metal
Mobility μ	Drift velocity per unit electric field
Carrier lifetime τ	Average time a minority carrier exists before recombining

EXAM CORNER • Try yourself (answers in brackets)

- Find n_i ratio for Ge when T rises from 300 K to 350 K (E_g = 0.67 eV). [≈ 6]
- A Si bar has $n = 5\times10^{21}$ m^{-3}, μ_e = 0.13. Find σ. [104 (Ωm)$^{-1}$]
- Hall coefficient of a sample is -8×10^{-5} m³/C. Find n and the carrier type. [7.8×10^{22} m^{-3}; electrons]
- Find V_H for I = 5 mA, B = 0.2 T, t = 0.2 mm, $n = 5\times10^{21}$ m^{-3}. [6.3 mV]

KEY IDEA • Problem-solving recipe

- **1.** Identify the type of material (intrinsic or n/p) and the temperature.
- **2.** Convert units: cm^{-3} → m^{-3} (×10^6), mm → m, eV → J (×1.6×10^{-19}); at 300 K kT = 0.0259 eV.
- **3.** Majority carrier ≈ doping level; get the minority carrier from $np = n_i^2$.
- **4.** Use $\sigma = e(n\mu_e + p\mu_h)$; locate E_F with $kT \ln(n/n_i)$ from E_i.

REMEMBER • Common mistakes

- Writing E_F at mid-gap for doped material (that is only for intrinsic).
- Forgetting that holes move along E while electrons move against E.
- Using the wrong sign for R_H of n-type material.
- Mixing units of mobility (cm²/Vs vs m²/Vs; 1 m²/Vs = 10^4 cm²/Vs).

8 Formula Sheet and Exam Corner

Topic	Key formulae
Intrinsic carriers	$n = N_c e^{-(E_c-E_F)/kT}$; $p = N_v e^{-(E_F-E_v)/kT}$; $np = n_i^2 = N_c N_v e^{-E_g/kT}$
Intrinsic E_F	$(E_c+E_v)/2 + (3kT/4)\ln(m_h^*/m_e^*)$
Conductivity	$\sigma = e(n\mu_e + p\mu_h)$; intrinsic: $\propto e^{-E_g/2kT}$
Extrinsic E_F	n-type: $(E_c+E_d)/2$ at 0 K; p-type: $(E_v+E_a)/2$ at 0 K
Currents	$J = e(n\mu_n + p\mu_p)E + e(D_n dn/dx - D_p dp/dx)$; $D/\mu = kT/e$
Hall	$V_H = IB/(net)$; $R_H = -1/ne$; $\mu = \lvert R_H\rvert\sigma$

EXAM CORNER • Short questions (2–3 marks)

- How are energy bands formed in solids?
- Distinguish between conductors, semiconductors and insulators using band diagrams.
- What is an intrinsic semiconductor? Where is its Fermi level?
- What are donor and acceptor impurities? Give examples.
- State the law of mass action.
- Define drift and diffusion currents.
- State Einstein's relation.
- Define Hall coefficient and write its expression.

EXAM CORNER • Long questions (10 marks)

- Derive expressions for the electron and hole concentrations in an intrinsic semiconductor and locate its Fermi level.
- Obtain an expression for the electrical conductivity of an intrinsic semiconductor and explain its temperature dependence.
- Derive the carrier concentration in n-type and p-type semiconductors. How does the Fermi level depend on doping and temperature?
- Explain drift and diffusion currents and derive Einstein's relation.
- Explain the Hall effect, derive the Hall coefficient and list its applications.

EXAM CORNER • Quick MCQs

- 1. Fermi level of an n-type semiconductor at 0 K lies (a) mid-gap (b) between E_d and E_c (c) below E_v.
- 2. Hall coefficient of a p-type semiconductor is (a) $-1/ne$ (b) $+1/pe$ (c) zero.
- 3. Resistivity of a semiconductor with rising T (a) increases (b) decreases (c) constant.
- 4. D/μ equals (a) kT/e (b) e/kT (c) kTe.
- **Answers:** 1-b, 2-b, 3-b, 4-a.

QUICK SUMMARY

- Atomic levels broaden into bands; the gap size separates conductors, semiconductors and insulators.
- Intrinsic: $n = p = n_i$, E_F at mid-gap, $\sigma \propto e^{-E_g/2kT}$. Extrinsic: doping fixes the majority carrier and moves E_F toward E_c (n) or E_v (p).
- Drift (field) and diffusion (gradient) carry current; $D/\mu = kT/e$. The Hall effect gives carrier type, density and mobility.